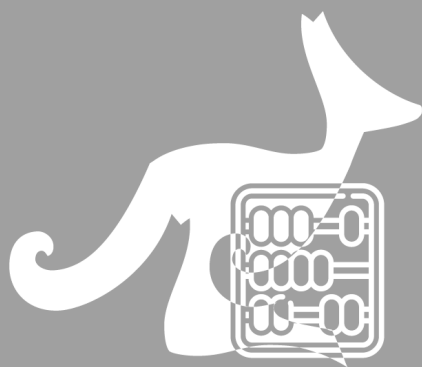




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a

2B pencil.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

Good Luck!

预祝考试顺利

**Grade 11
&
Grade 12**

3 points

1. What is the value of

计算下面算式的值

$$\frac{7777^2}{5555 \times 2222} ?$$

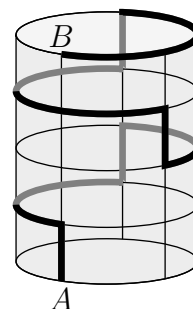
- (A) 1 (B) $\frac{7}{10}$ (C) $\frac{49}{10}$ (D) $\frac{77}{110}$ (E) 49

2. Giulia rolls five dice. She rolls 19 points in total. What is the maximum number of sixes she could have rolled?

Giulia 投掷了五枚骰子，点数之和等于 19。最多有多少枚骰子点数是 6？

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

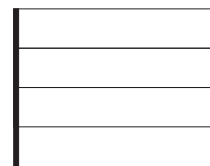
3. A cylindrical can has height 15 cm and the perimeter of its circular base is 30 cm. An ant walks from point A on the base to point B on the roof. Its path is either vertically upwards or horizontally along circular arcs around the can. Its path is shown with a thicker line (black for the path on the front of the can and grey at the back). What is the length, in cm, of the ant's path?



圆柱形罐子的高度为 15 厘米，底面圆形的周长为 30 厘米。一只蚂蚁从底面上的 A 点爬到顶面上的 B 点，爬行方向要么垂直向上要么沿着罐子的圆弧水平前进。如图所示，粗线（黑色为罐子正面的路径，灰色为罐子背面的路径）表示蚂蚁的爬行路径。蚂蚁爬行路径的总长度是多少厘米？

- (A) 45 (B) 55 (C) 60
(D) 65 (E) 75

4. Emma has four different coloured pens. She wants to colour the three-striped rectangular flag shown in the diagram so that each stripe is a single colour and no two adjacent stripes are the same colour. In how many ways can she do this?



Emma 有四支不同颜色的笔。她想给图中所示的三横条长方形旗帜涂色，每个横条都涂一种颜色，且相邻条纹的颜色不同。她能用多少种方式给旗帜涂色？

- (A) 24 (B) 27 (C) 32 (D) 36 (E) 64

5. We call a positive integer n *two-prime*, if it has exactly three different divisors, namely 1, 2 and n itself. How many different two-prime integers are there?

如果一个正整数 n 恰好有三个不同的因子，即 1、2 和 n 本身，则称 n 为“二素数”。一共有多少个不同的“二素数”？

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

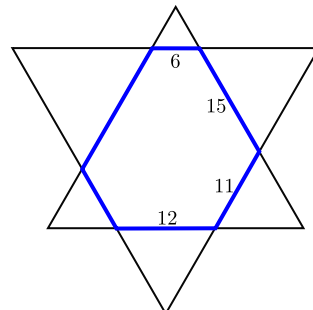
6. How many pairs of positive integers x and y satisfy the equation $x + 2y = 2^{10}$?

有多少对正整数 x 和 y 满足方程 $x + 2y = 2^{10}$?

- (A) $2^9 - 1$ (B) 2^9 (C) $2^9 + 1$ (D) $2^9 + 2$ (E) 0

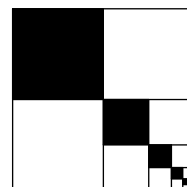
7. Two equilateral triangles are put together to form a hexagon with their opposite sides parallel. We know the length of four sides of this hexagon, as shown in the diagram. What is the perimeter of the hexagon?

将两个等边三角形放在一起形成一个六边形，六边形的对边互相平行。如图所示，六边形的四条边长已经标出。这个六边形的周长是多少？



- (A) 64
(B) 66
(C) 68
(D) 70
(E) 72

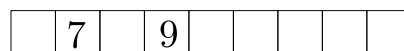
8. A square with area 84 is divided into four squares. The upper left square is coloured black. The lower right square is again divided into four squares, and so on. The process is repeated an infinite number of times. What is the total area that is coloured black?



将面积为 84 的正方形分成四个小正方形后，将左上角的小正方形涂为黑色。然后将右下角的小正方形再次分成四个更小的正方形，以此类推，并将这个过程重复无数次。涂成黑色的正方形的总面积是多少？

- (A) 24 (B) 28 (C) 31 (D) 35 (E) 42

9. Each of the integers from 1 to 9 is to be placed in one of the 9



boxes in the picture so that any three numbers in consecutive

boxes add to a multiple of 3. The numbers 7 and 9 have already been placed.

In how many different ways can the remaining boxes be filled?

将数字 1 到 9 填入图中 9 个方格的任一方格内，使得任意三个连续方格中的数字之和为 3 的倍数。数字 7 和 9 已经填入方格。一共有多少种不同方式来放置剩余数字？

- (A) 9 (B) 12 (C) 15 (D) 18 (E) 24

10. What is the units digit of the product $(5^5 + 1)(5^{10} + 1)(5^{15} + 1)$?

$(5^5 + 1)(5^{10} + 1)(5^{15} + 1)$ 之积的个位数是多少？

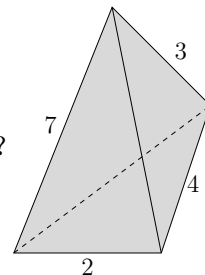
- (A) 1 (B) 2 (C) 4 (D) 5 (E) 6

4 points

11. A triangular pyramid has edges of integer length. Four of these lengths are as shown in the diagram. What is the sum of the lengths of the other two edges?

三棱锥的边长均为整数，其中四条边长如图所示。剩余两条边的长度之和是多少？

- (A) 9 (B) 10
(C) 11 (D) 12
(E) 13



12. For a positive integer n , $n!$ is defined as the product of all integers from 1 to n . For example, $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$. What is the sum of the digits of N if $N! = 6! \cdot 7!$?

对于正整数 n , $n!$ 定义为从 1 到 n 之间的所有整数之积。例如, $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$ 。如果 $N! = 6! \cdot 7!$, N 的各个数位上的数字之和是多少?

- (A) 1 (B) 2 (C) 4 (D) 8 (E) 9

13. The graphs of the functions $y = x^3 + 3x^2 + ax + 2a + 4$ all pass through the same point, no matter what value of a is chosen. What is the sum of the coordinates of that point?

无论 a 取何值, 函数 $y = x^3 + 3x^2 + ax + 2a + 4$ 的图像都会经过同一个点。这个点的横纵坐标之和是多少?

- (A) 2 (B) 4 (C) 7 (D) 8
(E) none of the previous 以上都不正确

14. We are given five numbers a_1, a_2, a_3, a_4, a_5 whose sum is S . For each k , $1 \leq k \leq 5$, we know that $a_k = k + S$. What is the value of S ?

五个数 a_1, a_2, a_3, a_4, a_5 之和为 S , 对于每个 k ($1 \leq k \leq 5$), 满足 $a_k = k + S$, S 的值是多少?

- (A) $\frac{15}{4}$ (B) $-\frac{15}{4}$ (C) -15 (D) 15
(E) none of the previous 以上都不正确

15. How many pairs of integers m and n satisfy the inequality $|2m - 2023| + |2n - m| \leq 1$?

有多少个 m 和 n 的整数对满足不等式 $|2m - 2023| + |2n - m| \leq 1$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

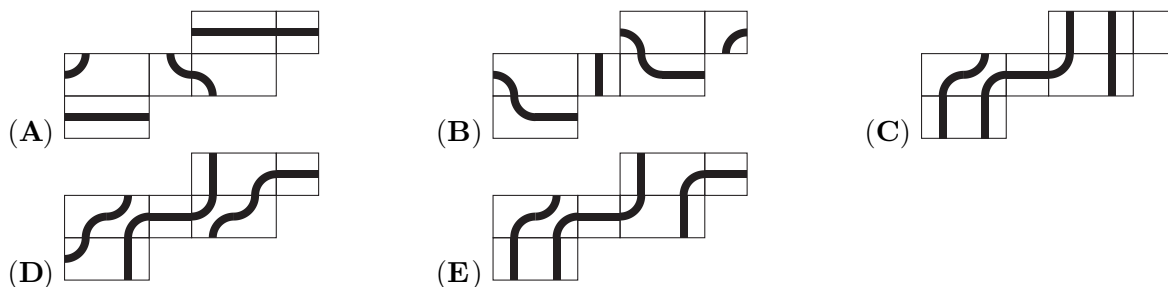
16. There are 23 animals sitting in a row at the cinema. Each animal is either a beaver or a kangaroo. Everyone has at least one neighbour who is a kangaroo. What is the largest possible number of beavers in the row?

电影院里有 23 只动物坐成一排, 它们要么是海狸要么是袋鼠。每只动物至少与一只袋鼠相邻。这一排最多可能有多少只海狸?

- (A) 7 (B) 8 (C) 10 (D) 11 (E) 12

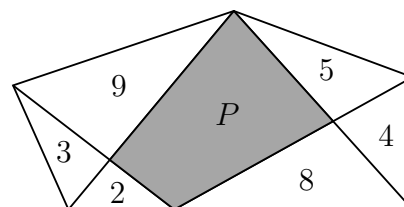
17. The number 5^{56} can be written in the form n^n for some integer n . What is the value of n ?
对于某一整数 n , 数字 5^{56} 可以写成 n^n 的形式, n 的值是多少?
- (A) 5^{30} (B) 5^6 (C) 5^5 (D) 30 (E) 11

18. Leon has drawn a closed path on a rectangular prism. Which net could show his path?
Leon 在长方体上画了一条闭合路径。下面哪张展开图可能是他画的路径?



19. A pentagon is dissected into smaller parts, as shown. The numbers inside the triangles indicate their areas. What is the area P of the shaded quadrilateral?

如图所示, 一个五边形被分成几个部分。三角形里面的数字表示这个三角形的面积。阴影四边形的面积 P 是多少?



- (A) 15 (B) $\frac{31}{2}$ (C) 16 (D) 17 (E) 18
20. How many integers are factors of $2^{20}3^{23}$ but are not factors of $2^{10}3^{20}$?
有多少个整数是 $2^{20}3^{23}$ 的因数, 但不是 $2^{10}3^{20}$ 的因数?
- (A) 13 (B) 30 (C) 273 (D) 460 (E) none of the previous 以上都不正确

5 points

21. Two functions f and g on $\mathbb{R}$ satisfy the system of equations

$$f(x) + 2g(1-x) = x^2 \text{ and } f(1-x) - g(x) = x^2. \text{ What is } f?$$

两个函数 f 和 g 位于 $\mathbb{R}$ 上, 且满足方程组

$$f(x) + 2g(1-x) = x^2 \text{ 和 } f(1-x) - g(x) = x^2, \text{ 求 } f.$$

- (A) $x^2 - \frac{4}{3}x + \frac{2}{3}$ (B) $x^2 + \frac{4}{3}x + \frac{2}{3}$ (C) $-x^2 - \frac{4}{3}x + \frac{2}{3}$
(D) $x^2 - 4x + 5$ (E) there are no such functions 不存在这样的函数

22. In a bouldering competition, 13 climbers compete in three categories. The score of each competitor is the product of their rankings in the three categories. For example, if one is 4th, 3rd and 6th, their final score is $4 \cdot 3 \cdot 6 = 72$. The higher your score, the lower your overall ranking. Hannah ranks 1st in two of the categories. What is her lowest possible overall ranking?

在一次攀岩比赛中，13 名运动员进行三个类别的比赛。每个参赛者的最终得分是他们在三个类别中排名的乘积。例如，一名运动员在三个类别的排名是第 4 名、第 3 名和第 6 名，那么他的最终得分为 $4 \cdot 3 \cdot 6 = 72$ 。参赛者的分数越高，总体排名就越低。Hannah 在其中两个类别中均排名第一，她的总体排名最低可能是多少？

- (A) 2nd 第二名 (B) 3rd 第三名 (C) 4th 第四名 (D) 5th 第五名 (E) 6th 第六名

23. A spiral of consecutive numbers is created, as shown, starting with 1. When the pattern of the spiral is continued, in which arrangement will the numbers 625, 626 and 627 appear?

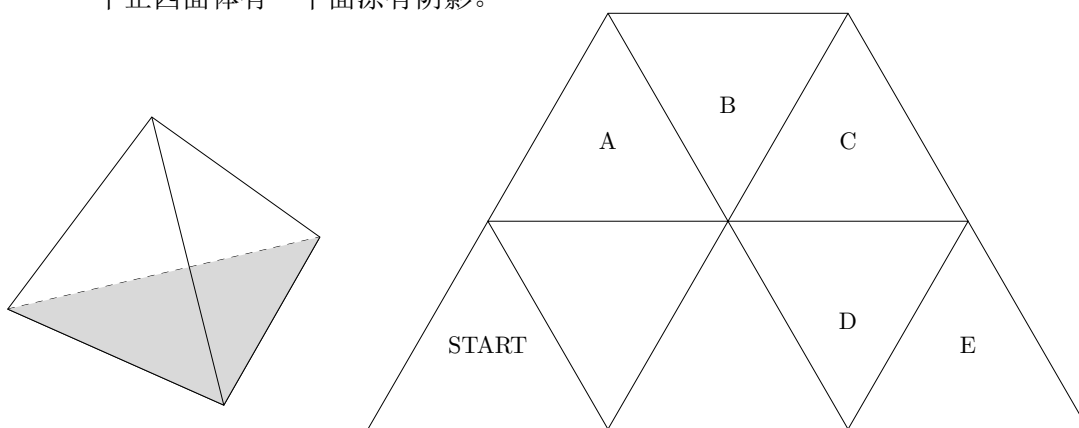
从数字 1 开始，呈螺旋形排列后面的连续数字，如图所示。数字螺旋上升过程中，625、626 和 627 将呈现以下哪种排列方式？

$\vdots$	10	$\rightarrow$	11	$\rightarrow$	12	$\rightarrow$	13
$\uparrow$	$\uparrow$						$\downarrow$
24	9		2	$\rightarrow$	3		14
$\uparrow$	$\uparrow$		$\uparrow$		$\downarrow$		$\downarrow$
23	8		1		4		15
$\uparrow$	$\uparrow$				$\downarrow$		$\downarrow$
22	7	$\leftarrow$	6	$\leftarrow$	5		16
$\uparrow$							$\downarrow$
21	$\leftarrow$	20	$\leftarrow$	19	$\leftarrow$	18	$\leftarrow$
							17

- | | | |
|--|--|--|
| <p style="text-align: center;">627</p> <p style="text-align: center;">$\uparrow$</p> <p>(A) 626</p> <p style="text-align: center;">$\uparrow$</p> <p style="text-align: center;">625</p> | <p style="text-align: center;">626 $\rightarrow$ 627</p> <p>(B) $\uparrow$</p> <p style="text-align: center;">625</p> | <p>(C) 625 $\rightarrow$ 626 $\rightarrow$ 627</p> |
| <p style="text-align: center;">625 $\rightarrow$ 626</p> <p>(D) $\downarrow$</p> <p style="text-align: center;">627</p> | <p style="text-align: center;">625</p> <p style="text-align: center;">$\downarrow$</p> <p>(E) 626</p> <p style="text-align: center;">$\downarrow$</p> <p style="text-align: center;">627</p> | |

24. A block in the shape of a regular tetrahedron has one face shaded.

一个正四面体有一个面涂有阴影。



The shaded face of the block is placed on the board on the triangle labelled START. The block is then rolled from one triangle to the next by rotating it about one edge. On which triangle will the block stand for the first time again on its shaded face?

将这个立方体的阴影面放置在上木板标有 START 的三角形上，然后立方体沿一条边进行翻转，从木板上的一个三角形滚动到下一个三角形。立方体的阴影面会再次首先滚动到哪个三角形上？

- (A) A (B) B (C) C (D) D (E) E

25. Part of the fifth degree polynomial shown cannot be seen because of an inkblot. It is known that all five roots of the polynomial are integers. What is the highest power of $x - 1$ that divides the polynomial?

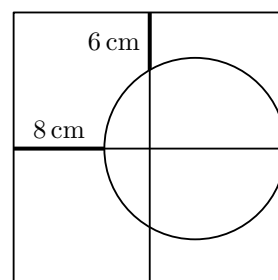
$$x^5 - 11x^4 + \text{[inkblot]} - 7$$

如图所示，一个五次多项式有一部分被墨水遮挡了。已知多项式的五个根都是整数，将该多项式进行分解后得到的 $x - 1$ 的最高次项是什么？

- (A) $(x - 1)^1$ (B) $(x - 1)^2$ (C) $(x - 1)^3$ (D) $(x - 1)^4$ (E) $(x - 1)^5$

26. The large square in the diagram is dissected into four smaller squares. The circle touches the right hand side of the square at its midpoint. What is the side-length of the large square? Note that the diagram is not drawn to scale.

图中的大正方形被分成四个小正方形。圆与正方形右侧边的中点相切。大正方形的边长是多少？请注意，图形未按照比例绘制。



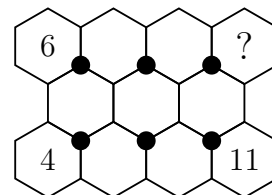
- (A) 18 cm (B) 20 cm
(C) 24 cm (D) 28 cm
(E) 30 cm

27. What is the greatest common divisor of all numbers of the form $n^3(n+1)^3(n+2)^3(n+3)^3(n+4)^3$, where n is a non-zero natural number?

以 $n^3(n+1)^3(n+2)^3(n+3)^3(n+4)^3$ 形式表示的所有数字的最大公约数是多少? 其中 n 是非零自然数。

- (A) $2^9 3^3$ (B) $2^3 3^3 5^3$ (C) $2^6 3^3 5^3$ (D) $2^8 3^2 5^3$ (E) $2^9 3^3 5^3$

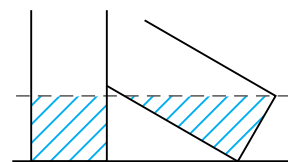
28. The numbers from 1 to 11 are to be placed in the hexagons so that the sum of the three numbers around each of the six black dots is the same. Three of the numbers have already been placed. What number will be placed in the hexagon with a question mark?



要将数字 1 到 11 放在右侧六边形中, 使得每个黑点周围的三个数字之和都相同。已经放好了三个数字, 带问号的六边形中会放置哪个数字?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

29. Two identical cylindrical water tanks contain the same amount of water. One cylinder is standing upright, and the other is leaning against it, and the water level in each of them is the same as in the picture. The bottom of each of the cylinders is a circle with area $3\pi m^2$. How much water does each tank contain?



两个完全相同的圆柱形水箱中装有相同体积的水。一个水箱直立放置, 另一个斜靠在这个水箱上面, 两个水箱的水位相同, 如图所示。两个圆柱形水箱的底部都是一个面积为 $3\pi m^2$ 的圆。每个水箱里装有多少水?

- (A) $3\sqrt{3}\pi m^3$ (B) $6\pi m^3$ (C) $9\pi m^3$ (D) $\frac{3\pi}{4} m^3$
(E) it's impossible to determine from the information given 根据已知信息无法确定

30. The product of six consecutive numbers is a 12-digit number of the form

$$abb\ cdd\ cdd\ abb,$$

where the digits a , b , c and d are themselves four consecutive numbers in some order. What is the value of the digit d ?

六个连续数字之积是一个 12 位数, 可以表示为以下形式

$$abb\ cdd\ cdd\ abb,$$

其中数字 a 、 b 、 c 和 d 以某种顺序排列是四个连续数字。数字 d 的值是多少?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Student

3 points

1 (Greece). What is the value of

$$\frac{7777^2}{5555 \times 2222}?$$

- (A) 1 (B) $\frac{7}{10}$ (C) $\frac{49}{10}$ (D) $\frac{77}{110}$ (E) 49

SOLUTION: We do not attempt the multiplications but instead work with simplifications.

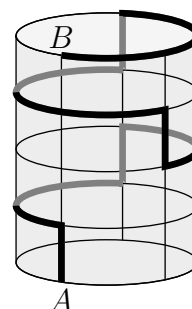
$$\frac{7777^2}{5555 \times 2222} = \frac{7^2 \times 1111^2}{5 \times 1111 \times 2 \times 1111} = \frac{7^2}{5 \times 2} = \frac{49}{10}$$

2 (Germany). Giulia rolls five dice. She rolls 19 points in total. What is the maximum number of sixes she could have rolled?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

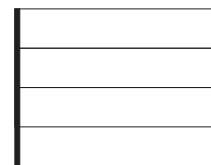
SOLUTION: If Giulia rolls four sixes, she has at least 24 points. She cannot roll 3 sixes, as $3 \cdot 6 = 18$, and she must roll at least one with each of the other two dice, and $18 + 1 + 1 = 20$ in total. She can, however, roll two sixes. For instance, she can roll $6 + 6 + 5 + 1 + 1 = 19$. The correct answer is therefore 2.3 (Greece). A cylindrical can has height 15 cm and the perimeter of its circular base is 30 cm. An ant walks from point A on the base to point B on the roof. Its path is either vertically upwards or horizontally along circular arcs around the can. Its path is shown with a thicker line (black for the path on the front of the can and grey at the back). What is the length, in cm, of the ant's path?

- (A) 45 (B) 55 (C) 60
(D) 65 (E) 75

SOLUTION: We do not calculate the individual parts of the path, but we look at them as a whole. The vertical parts add up to the height of the can, so it is 15 cm. The horizontal paths, if projected to the ground, can be seen to be two full base circles, so their length is $2 \times 30 = 60$ cm. The grand total is $15 + 60 = 75$ cm.

4 (Slovenia). Emma has four different coloured pens. She wants to colour the three-striped rectangular flag shown in the diagram so that each stripe is a single colour and no two adjacent stripes are the same colour. In how many ways can she do this?

- (A) 24 (B) 27 (C) 32 (D) 36 (E) 64

SOLUTION: The top stripe can be in any of the four colors. Each of these can be combined with three of the colors, as no two adjoining stripes can be of the same color. The bottom stripe can again be any of the three colors not in the middle. This means that there are a total of $4 \cdot 3 \cdot 3 = 36$ possible flags.5 (Austria). We call a positive integer n two-prime, if it has exactly three different divisors, namely 1, 2 and n itself. How many different two-prime integers are there?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

SOLUTION: The only two-prime number is $2^2 = 4$. Any number $n = 2 \cdot k$ is divisible by 2 and k (as well as 1 and n), and therefore $k = 2$ must hold for a two-prime number.

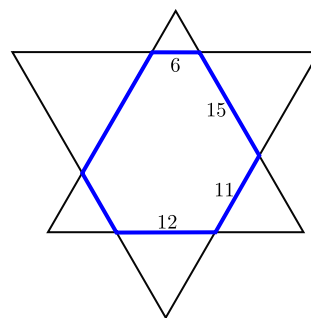
6 (Montenegro). How many pairs of positive integers x and y satisfy the equation $x + 2y = 2^{10}$?

- (A) $2^9 - 1$ (B) 2^9 (C) $2^9 + 1$ (D) $2^9 + 2$ (E) 0

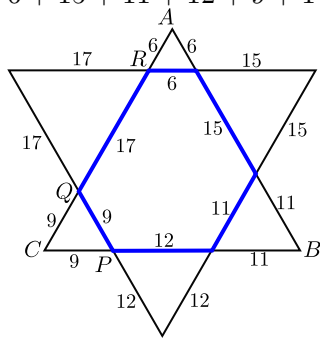
SOLUTION: If the value of y is any of the numbers from 1 to $2^9 - 1$, there is a solution in positive integers for $x = 2^{10} - 2y$, as we then have $2y < 2^{10}$. For all other values of y , there is no appropriate value of x . We therefore have a total of $2^9 - 1$ pairs of integers satisfying the equation.

7 (Mexico). Two equilateral triangles are put together to form a hexagon with their opposite sides parallel. We know the length of four sides of this hexagon, as shown in the diagram. What is the perimeter of the hexagon?

- (A) 64
(B) 66
(C) 68
(D) 70
(E) 72

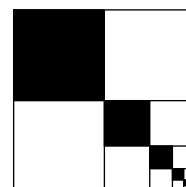


SOLUTION: Because the sides are parallel, all the tips of the star are equilateral triangles and hence the side AB of triangle ABC in the picture is $6 + 15 + 11 = 32$ long, but then the length of BC is also 32, so that $PC = 32 - (12 + 11) = 9$ and $QR = 32 - (9 + 6) = 17$. The perimeter is $6 + 15 + 11 + 12 + 9 + 17 = 70$.



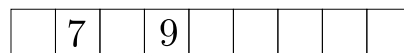
8 (Netherlands). A square with area 84 is divided into four squares. The upper left square is coloured black. The lower right square is again divided into four squares, and so on. The process is repeated an infinite number of times. What is the total area that is coloured black?

- (A) 24 (B) 28 (C) 31 (D) 35 (E) 42



SOLUTION: At each stage, $\frac{1}{3}$ of the smaller squares are coloured black so in total $\frac{1}{3}$ of the original square is black giving a black area of 28.

9 (Mexico). Each of the integers from 1 to 9 is to be placed in one of the 9 boxes in the picture so that any three numbers in consecutive boxes add to a multiple of 3. The numbers 7 and 9 have already been placed. In how many different ways can the remaining boxes be filled?



- (A) 9 (B) 12 (C) 15 (D) 18 (E) 24

SOLUTION: Noting that $7 = 2 \cdot 3 + 1$ and $9 = 3 \cdot 3$, the number between 7 and 9 must be of the form $3x + 2$ in order for the sum of the three to be divisible by 3. This gives us the options 2, 5 and 8 for this square. The left square must be filled with a number of the form $3x$ for the analogous reason, and this gives the options 3 and 6. We therefore have a total of $3 \cdot 2 = 6$ possible ways to fill the two squares next to the digit 7. Next to the 9, we again need a digit of the form $3x + 1$, and therefore 4 or 1, and next to that, a digit of the form $3x + 2$, of which we still have two digits not yet used. This gives us a total of $6 \cdot 2 \cdot 2 = 24$ ways to fill the 6 leftmost squares. This forces a single option for each of the remaining squares, as these must be of the form $3x$, $3x + 1$ and $3x + 2$, respectively, and there is only one digit of each type left to place. This means that there is a total of 24 options to distribute the digits in a way that fulfills the conditions.

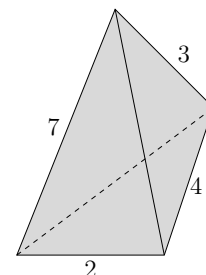
10 (United Kingdom). What is the units digit of the product $(5^5 + 1)(5^{10} + 1)(5^{15} + 1)$?

- (A) 1 (B) 2 (C) 4 (D) 5 (E) 6

SOLUTION: The units digit of 5^n is 5 for all positive integers n . Therefore the units digit of $5^n + 1$ is 6. Hence the units digit of the product is the same as the units digit of 6^3 which is 6.

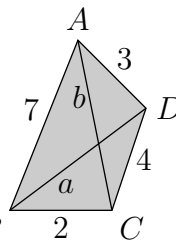
4 points

11 (Greece). A triangular pyramid has edges of integer length. Four of these lengths are as shown in the diagram. What is the sum of the lengths of the other two edges?



- (A) 9 (B) 10
(C) 11 (D) 12
 (E) 13

SOLUTION: We use the triangle inequality several times. From the triangle ABD we have $7 - 3 < a < 7 + 3$, so that (the integer) $a \in \{5, 6, 7, 8, 9\}$. Also from the triangle BCD we have $4 - 2 < a < 4 + 2$, so $a \in \{3, 4, 5\}$. It follows that a is in the intersection of the two set, so $a = 5$. Similarly from the triangles ABC and ACD we have $7 - 2 < b < 7 + 2$ and $4 - 1 < b < 4 + 3$. These give $b \in \{6, 7, 8\}$



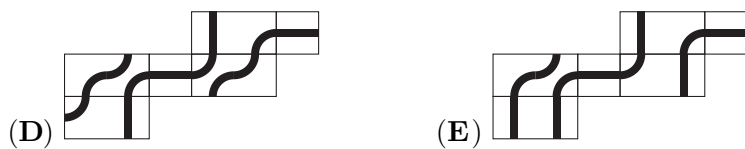
and $b \in \{4, 5, 6\}$, so $b = 6$. Finally, $a + b = 5 + 6 = 11$.

12 (Turkey). For a positive integer n , $n!$ is defined as the product of all integers from 1 to n . For example, $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$. What is the sum of the digits of N if $N! = 6! \cdot 7!$?

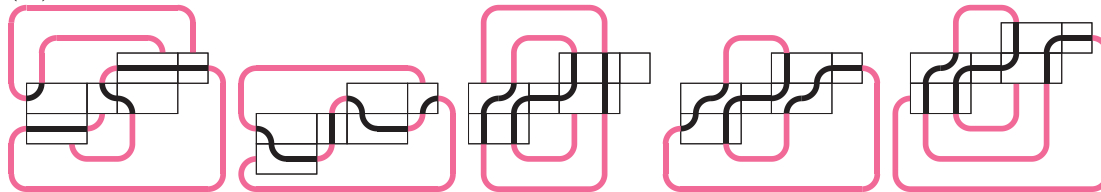
- (A) 1 (B) 2 (C) 4 (D) 8 (E) 9

SOLUTION: $7! \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 7! \cdot 3 \cdot 2 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 7! \cdot 8 \cdot 9 \cdot 10 = 10!$

13 (Greece). The graphs of the functions $y = x^3 + 3x^2 + ax + 2a + 4$ all pass through the same



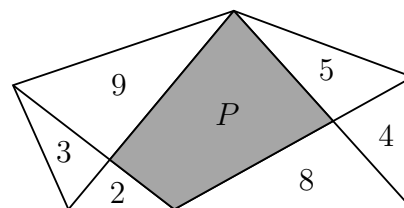
SOLUTION: The nets given can be folded into solid uniquely. Following the path one can see that only (D) is closed.



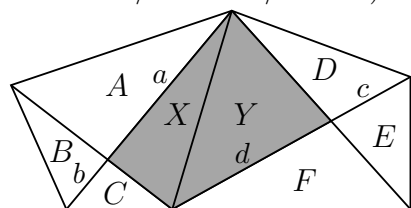
Connections in magenta show which pieces are glued together (i.e. belong to the same path).

19 (Greece). A pentagon is dissected into smaller parts, as shown. The numbers inside the triangles indicate their areas. What is the area P of the shaded quadrilateral?

- (A) 15 (B) $\frac{31}{2}$ (C) 16 (D) 17
(E) 18



SOLUTION: Draw the diagonal across the shaded area, as shown. Areas A and B have the same altitude and $A = 3B$ so their bases satisfy $a = 3b$. It follows that $X = 3C = 6$. Similarly $F = 2E$ so $d = 2c$ and so $Y = 2D = 10$. We conclude that $P = X + Y = 16$. (Note that, more generally, what this is really saying is that $AC = BX$ and $DF = EY$. So the required area is $X + Y = AC/B + DF/E$, here $9 \times 2/3 + 5 \times 8/4 = 16$.)



20 (Greece). How many integers are factors of $2^{20}3^{23}$ but are not factors of $2^{10}3^{20}$?

- (A) 13 (B) 30 (C) 273 (D) 460
(E) none of the previous

SOLUTION: Note that the first number, $2^{20}3^{23}$, is a multiple of the second, $2^{10}3^{20}$. It follows that each divisor of the second number is also a divisor of the first. The total number of divisors of the first number is $(20 + 1)(23 + 1) = 21 \cdot 24$ and of the second is $(10 + 1)(20 + 1) = 11 \cdot 21$. So the difference of the two, namely $21 \cdot 24 - 11 \cdot 21 = 21(24 - 11) = 21 \cdot 13 = 273$ is the required answer.

5 points

21 (Greece). Two functions f and g on $\mathbb{R}$ satisfy the system of equations $f(x) + 2g(1 - x) = x^2$ and $f(1 - x) - g(x) = x^2$. What is f ?

- (A) $x^2 - \frac{4}{3}x + \frac{2}{3}$ (B) $x^2 + \frac{4}{3}x + \frac{2}{3}$ (C) $-x^2 - \frac{4}{3}x + \frac{2}{3}$
(D) $x^2 - 4x + 5$ (E) there are no such functions

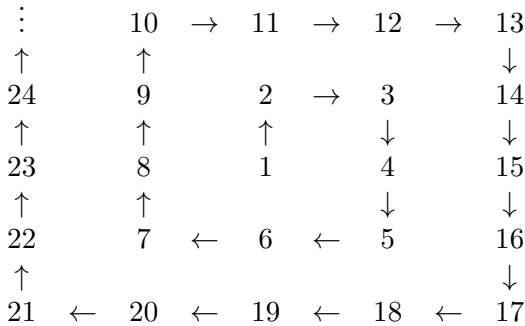
SOLUTION: Putting $1 - x$ in the place of x , the second equation becomes $f(x) - g(1 - x) = (1 - x)^2 = 1 - 2x + x^2$. Solving the system of the first equation and this last in terms of $f(x)$ and $g(1 - x)$, we find $f(x) = x^2 - \frac{4}{3}x + \frac{2}{3}$ and $g(x) = -\frac{2}{3}x + \frac{1}{3}$.

22 (Germany). In a bouldering competition, 13 climbers compete in three categories. The score of each competitor is the product of their rankings in the three categories. For example, if one is 4th, 3rd and 6th, their final score is $4 \cdot 3 \cdot 6 = 72$. The higher your score, the lower your overall ranking. Hannah ranks 1st in two of the categories. What is her lowest possible overall ranking?

- (A) 2nd (B) 3rd (C) 4th (D) 5th (E) 6th

SOLUTION: We show a way that Hanna can be third. Hanna's points can be $1 \cdot 1 \cdot 13 = 13$. One competitor's points in this case can be $2 \cdot 2 \cdot 2 = 8$, while another competitor simultaneously has $3 \cdot 3 \cdot 1 = 9$ points. We see that it is possible for Hanna to place third. We will now show that there cannot be more than two competitors ahead of Hanna. Hanna's points are at most 13. Let's say there are 3 competitors ahead of Hanna. Their points in the first competition are at least 2, 3 and 4, as are their points in the second competition. Their points in the third competition are at least 1, 2 and 3. The product of the total points of these three contestants is therefore at least $2^3 \cdot 3^3 \cdot 4^2 = 3456$. If all three competitors had a maximum of 13 points, the product of their scores can be at most $13^3 = 2197$ points. This is a contradiction, and we see that there can be at most two competitors ahead of Hanna.

23 (Serbia). A spiral of consecutive numbers is created, as shown, starting with 1. When the pattern of the spiral is continued, in which arrangement will the numbers 625, 626 and 627 appear?



- | | | |
|--|--|--|
| <p style="text-align: center;">627
$\uparrow$
(A) 626
$\uparrow$
625</p> | <p style="text-align: center;">626 $\rightarrow$ 627
(B) $\uparrow$
625</p> | <p style="text-align: center;">(C) 625 $\rightarrow$ 626 $\rightarrow$ 627</p> |
| <p style="text-align: center;">(D) 625 $\rightarrow$ 626
$\downarrow$
625</p> | <p style="text-align: center;">(E) 625
$\downarrow$
626
$\downarrow$
627</p> | |

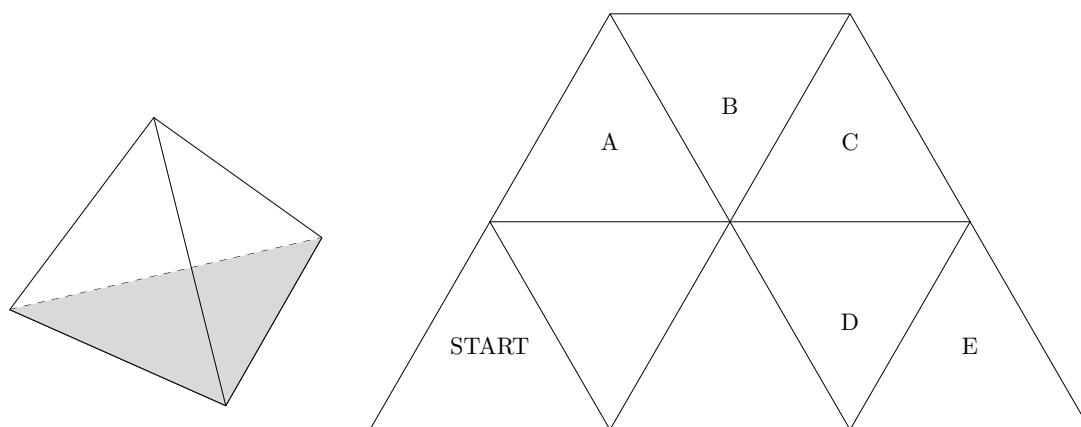
SOLUTION: If you start from 1 you must add 1 and then 1 to get to the first top right corner and then 2 and 2 to get to the bottom left corner, and so on, so that the numbers at the bottom left corners are always of the form $1 + 2n(2n + 1)$ (after adding $2n$ twice). Note that

$$(2n + 1)^2 = 1 + 2n(2n + 1) + 2n$$

which means that the squares of odd numbers will always be one step away from the top left corners of the spirals (since after adding $2n$ twice you must add $2n + 1$ to reach the top left corner). Now $626 = 25^2 + 1$ so 626 is at the top left corner which makes (B) the correct answer.

24 (Poland). A block in the shape of a regular tetrahedron has one face shaded.

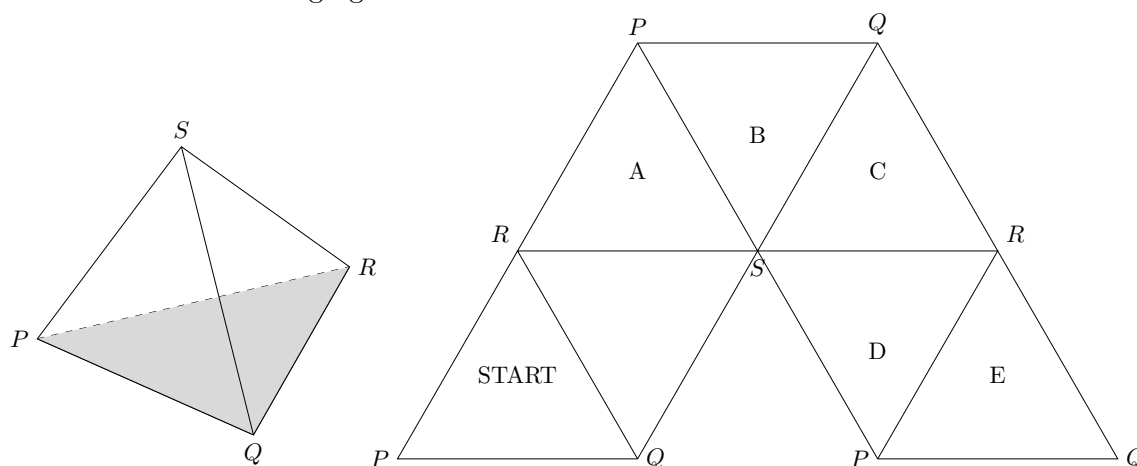
Student Finalized



The shaded face of the block is placed on the board on the triangle labelled START. The block is then rolled from one triangle to the next by rotating it about one edge. On which triangle will the block stand for the first time again on its shaded face?

- (A) A (B) B (C) C (D) D (E) E

SOLUTION: Name the vertices of the tetrahedron P, Q, R, S , with P, Q, R being the vertices of the triangle in the starting position. Then we can follow the path of the vertices through the movements as shown in the following figure:



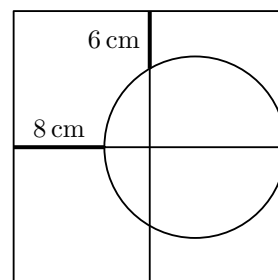
25 (Greece). Part of the fifth degree polynomial shown cannot be seen because of an inkblot. It is known that all five roots of the polynomial are integers. What is the highest power of $x - 1$ that divides the polynomial?

$$x^5 - 11x^4 + \text{[inkblot]} - 7$$

- (A) $(x - 1)^1$ (B) $(x - 1)^2$ (C) $(x - 1)^3$ (D) $(x - 1)^4$ (E) $(x - 1)^5$

SOLUTION: The product of the roots is 7 so the roots can only be of the form $\pm 1, \pm 7$ (but with exactly one 7 or -7). The sum of the roots is 11. The only way to get 11 as a sum of a ± 7 and (four) ± 1 's is $1+1+1+1+7=11$. So the roots are 1 with multiplicity 4 and 7, and so the polynomial is $(x-1)^4(x-7)$. (Note we can avoid Vieta's relationships since the students know that if a polynomial has roots $p, q, \dots$ then it is of the form $a(x-p)(x-q)\dots$ So we have $x^5 - 11x^4 + \dots - 7 = a(x-p)(x-q)\dots$ and simply compare coefficients).

26 (Germany). The large square in the diagram is dissected into four smaller squares. The circle touches the right hand side of the square at its midpoint. What is the side-length of the large square? Note that the diagram is not drawn to scale.



- (A) 18 cm
(B) 20 cm
(C) 24 cm
(D) 28 cm
(E) 30 cm

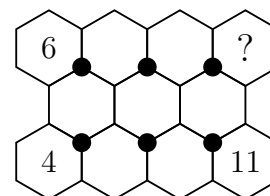
SOLUTION: Let $2a$ be the side length of the large square. Consider the right-angled triangle with hypotenuse the diameter of the circle that is shown and right angle at the lower tip of the segment of length 6 cm. For the altitude h on the hypotenuse we have (in cm) $h = a - 6$ and $h^2 = (a - 8) \cdot a$. So we have $(a - 6)^2 = (a - 8) \cdot a$ from which get $a = 9$.

27 (Greece). What is the greatest common divisor of all numbers of the form $n^3(n+1)^3(n+2)^3(n+3)^3(n+4)^3$, where n is a non-zero natural number?

- (A) $2^9 3^3$ (B) $2^3 3^3 5^3$ (C) $2^6 3^3 5^3$ (D) $2^8 3^2 5^3$ (E) $2^9 3^3 5^3$

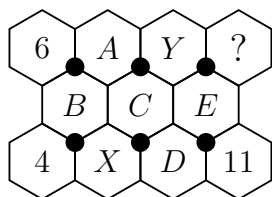
SOLUTION: For $n = 1$ we get the number $1^3 2^3 3^3 4^3 5^3 = 2^9 3^3 5^3$. So the greatest common divisor is at most $2^9 3^3 5^3$ or some submultiple of this. We show that it is actually this number itself. This is because $n(n+1)(n+2)(n+3)(n+4)$ is a run of five consecutive numbers so there are a) at least two multiples of 2 (and one of the two will be a multiple of 4), b) At least one multiple of 3 and c) at least one multiple of 5. So all the numbers are a multiple of at least $2 \times 4 \times 3 \times 5 = 2^3 \times 3 \times 5$. The cube of this is the required answer.

28 (Poland). The numbers from 1 to 11 are to be placed in the hexagons so that the sum of the three numbers around each of the six black dots is the same. Three of the numbers have already been placed. What number will be placed in the hexagon with a question mark?



- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

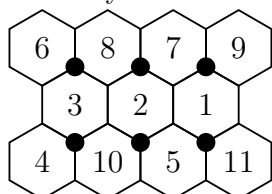
SOLUTION: Let's name the hexagons A, B, C, D, E, X and Y as in the figure



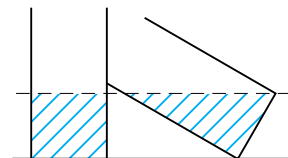
Considering the two leftmost dots we get $6 + A + B = 4 + B + X$, so $X = A + 2$.

Considering the two middle dots we get $Y = D + 2$, and finally, using the two rightmost dots we get $11 = ? + 2$, so $? = 9$.

The only solution is actually as follows.

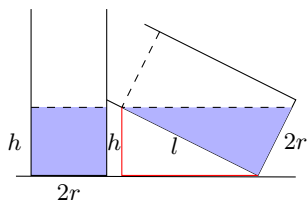


29 (Belarus). Two identical cylindrical water tanks contain the same amount of water. One cylinder is standing upright, and the other is leaning against it, and the water level in each of them is the same as in the picture. The bottom of each of the cylinders is a circle with area $3\pi m^2$. How much water does each tank contain?



- (A) $3\sqrt{3}\pi m^3$ (B) $6\pi m^3$ (C) $9\pi m^3$ (D) $\frac{3\pi}{4} m^3$

(E) it's impossible to determine from the information given



SOLUTION:

Note that $r = \sqrt{3}$ is the radius of the bottom of the cylinder. In the first (upright) cylinder we see the rectangle $2r \times h$ and in the second (leaning) cylinder we see the right triangle with the cathetuses $2r$ and ℓ . From symmetry considerations, it is clear that if we pour water into the upright cylinder up to the level ℓ , it will contain twice as much water as the second cylinder (and the first) is containing now. Thus, $\ell = 2h$.

Therefore the right triangle (pictured) with the cathetus h and hypotenuse ℓ has the angles 30° and 60° . From the right triangle with cathetuses $\ell = 2h$ and $2r$ we then obtain $h = \sqrt{3}r = \sqrt{3} \cdot \sqrt{3} = 3$. So, each cylinder contains $3 \cdot 3\pi = 9\pi m^3$ of water.

30 (Australia). The product of six consecutive numbers is a 12-digit number of the form

$$abb\ cdd\ cdd\ abb,$$

where the digits a , b , c and d are themselves four consecutive numbers in some order. What is the value of the digit d ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: Among the six consecutive factors in the product, three of them are even and at least one must be a multiple of 5. Hence the product is a multiple of 10 and so digit b must be zero. Since the digits are consecutive, it follows that a , c and d are equal to 1, 2 and 3, in some order. In particular,

$$a + c + d = 6.$$

Next notice that among the six consecutive factors, two of them must be multiples of 3. Hence the product is a multiple of 9 and so therefore is its digit sum. But the digit sum is also even, since each block of three digits occurs twice, so half the digit sum must also be a multiple of 9. Given that $b = 0$ and the remaining digits are at most 3, we deduce that

$$a + c + 2d = 9.$$

Subtracting the two equations shows that $d = 3$ (In fact, the product is $74 \times 75 \times 76 \times 77 \times 78 \times 79 = 200\ 133\ 133\ 200$, with $a = 2$ and $c = 1$.)